Multi-Channel Spectrogram Fusion with Digital Image Processing Enhancement for Convolutional Neural Network-Based Speech Emotion Recognition

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ABSTRACT

Speech Emotion Recognition (SER) is a fundamental problem in affective computing with wide applications in human–computer interaction, mental health monitoring, and social robotics [1],[2]. Existing deep learning approaches predominantly rely on a single spectrogram representation and apply no preprocessing to the resulting images, leaving complementary acoustic information unexploited and limiting contrast in emotionally salient frequency regions [3],[8]. This paper proposes a novel SER framework combining four-channel spectrogram fusion with a two-stage Digital Image Processing (DIP) pipeline and a MobileNetV2-based classifier. Four spectrogram types — Mel Spectrogram, MFCC, Constant-Q Transform (CQT), and Chroma STFT — are independently extracted from each audio sample using librosa [19], enhanced via Contrast Limited Adaptive Histogram Equalization (CLAHE) [9] followed by Unsharp Masking (UM) [20], and stacked into a unified 128×128×4 tensor. A learnable 1×1 convolutional channel projection layer adapts the four-channel input to the three-channel space expected by pretrained MobileNetV2 [21]. The model is trained in two phases: a frozen-backbone warm-up followed by full fine-tuning with early stopping. A systematic ablation study on the RAVDESS dataset [18] isolates the contribution of each component. Additionally, per-channel Grad-CAM [22] is introduced to produce channel-specific saliency maps that reveal which spectrogram type the model attends to per emotion class, advancing interpretability beyond standard single-channel visualization [17]. Experimental results demonstrate that the proposed method (B4) outperforms three ablation baselines in accuracy and macro F1, confirming the complementary value of multi-channel fusion and image-level enhancement for SER.

Keywords: speech emotion recognition; multi-channel spectrogram fusion; CLAHE; unsharp masking; MobileNetV2; Grad-CAM; transfer learning; affective computing; RAVDESS

I. INTRODUCTION

Automatic recognition of human emotional states from speech has become an increasingly critical capability in modern intelligent systems. Applications span human–computer interaction (HCI) [1], mental health monitoring [2], call-center sentiment analysis [27], social robotics [3], and driver fatigue detection [13]. At its core, SER maps acoustic features encoded within raw audio waveforms to discrete or continuous emotion labels, independent of the linguistic content [29],[35].

Deep learning — in particular CNN-based architectures applied to time-frequency spectrogram representations — has largely superseded classical feature engineering approaches [10],[15],[23]. Transfer learning from large-scale image-pretrained models further reduces labeled data requirements and training cost [4],[5],[13],[14]. However, two fundamental limitations persist. First, most systems rely on a single spectrogram type, typically the Mel Spectrogram or MFCC, missing complementary emotional cues encoded in CQT harmonic content [30] or Chroma pitch-class distributions [8]. Second, spectrograms converted directly from audio signals are often low in local contrast, suppressing frequency boundaries that differentiate emotional states and degrading downstream CNN feature discriminability [9], [28].

Related Work. CNN-based SER was pioneered by applying image classification backbones to Mel spectrograms [10], [11]. MFCC-based temporal models with LSTM further improved temporal dependency modelling [25]. Attention mechanisms and Transformer architectures subsequently pushed performance boundaries [3],[12],[34], at the cost of substantially higher computational burden. Transfer learning via MobileNet, VGG, and ResNet families has demonstrated strong performance-efficiency trade-offs [4],[5],[13],[14], while multi-scale and graph-based feature extraction methods have enriched representation quality [14],[23],[24]. Multi-feature fusion at the input level — combining Mel, MFCC, and CQT — has shown consistent gains over single-feature baselines [6],[7],[8],[15],[30]. Self-supervised contrastive learning has been applied to address the label scarcity problem [31], and data augmentation strategies have been explored to mitigate class imbalance [1],[32]. On the preprocessing side, CLAHE has improved classification on medical signal imagery [9] and radar spectrogram

classification [28], and Unsharp Masking has sharpened structural detail for image classifiers [20], yet neither technique has been systematically applied to SER spectrogram images. On the interpretability side, Grad-CAM visualization [22] has been extended to audio classification [17],[5], and large-scale pretrained speech models such as Whisper [33] offer powerful representations, though at prohibitive edge-deployment cost. Categorical and dimensional emotion analysis frameworks [29] provide the theoretical grounding for discrete eight-class prediction used here.

The present work addresses all identified gaps with four contributions: (1) a four-channel spectrogram fusion strategy (Mel+MFCC+CQT+Chroma) capturing complementary emotional cues; (2) a two-stage DIP pipeline (CLAHE + UM) enhancing each channel before fusion; (3) a 1×1 Conv channel projection enabling four-channel input to ImageNet-pretrained MobileNetV2; and (4) per-channel Grad-CAM providing interpretable, channel-level saliency per emotion class. Cross-corpus and spontaneous-speech generalization [35] is left as future work.

II. PROPOSED METHODOLOGY

A. Dataset

All experiments use the Ryerson Audio-Visual Database of Emotional Speech and Song (RAVDESS) [18], comprising 1,440 audio recordings from 24 professional actors (12 male, 12 female) under controlled studio conditions. Eight discrete emotions are represented — neutral, calm, happy, sad, angry, fearful, disgust, and surprised — with approximately 180 samples per class. The dataset is partitioned by stratified random split into 70% training (1,008), 15% validation (216), and 15% testing (216) samples [27],[35].

B. Multi-Channel Spectrogram Extraction

Each audio file is resampled to 22,050 Hz and padded or truncated to three seconds (66,150 samples). Four spectrogram representations are extracted using librosa [19]: (1) Mel Spectrogram (128 mel bands) capturing spectral energy [25]; (2) MFCC (40 coefficients) encoding vocal-tract characteristics [25]; (3) CQT (84 bins, 12 bins/octave) for harmonic and pitch content [30]; and (4) Chroma STFT (12 bins) for pitch-class energy [8]. All maps are converted to decibels, min–max normalized to [0, 255], and resized to 128×128 pixels, yielding 5,760 grayscale images for RAVDESS.

C. DIP Enhancement Pipeline

Each grayscale spectrogram undergoes two-stage preprocessing. Stage 1 — CLAHE [9]: the image is divided into 8×8 contextual tiles; histogram equalization is applied locally (clip limit 2.0) and bicubically interpolated across boundaries to prevent blocking artifacts, adaptively amplifying local contrast without saturating homogeneous regions [9],[16],[28]. Stage 2 — Unsharp Masking [20]: a Gaussian-blurred copy ($\sigma = 1.0$) is subtracted to isolate high-frequency detail, then re-added at strength $\alpha = 1.5$:

$$I_{sharp} = (1+\alpha)\cdot I_{CLAHE} - \alpha\cdot G_{\sigma}(I_{CLAHE}) \quad (1)$$

where G_σ denotes Gaussian blur with $\sigma = 1.0$ [20]. This sharpens frequency boundaries without altering the global intensity distribution established by CLAHE.

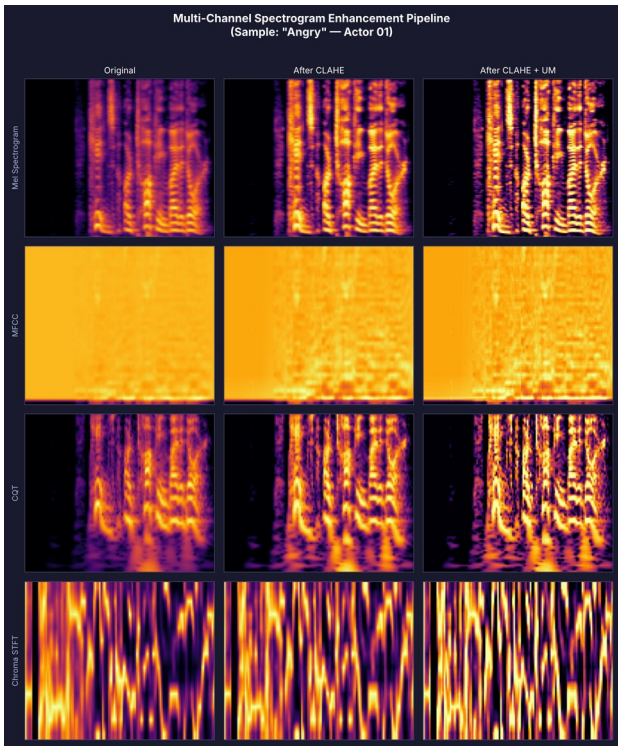


Figure 1. *Two-stage DIP enhancement per channel. Columns: original, after CLAHE, after CLAHE+UM.*

D. Four-Channel Fusion & Architecture

The four enhanced 128×128 images are stacked into a 128×128×4 tensor (normalized to [0,1]). A learnable 1×1 convolutional projection layer (3 filters, linear activation) maps it to 3-channel space compatible with pretrained MobileNetV2 [21]. MobileNetV2's depthwise separable convolutions and inverted residuals [21] provide efficiency for edge deployment [4],[13]. The backbone is followed by GAP, Dropout(0.3), Dense(128, ReLU), BN, Dropout(0.3), and a softmax output (N=8) for the eight RAVDESS classes [14].

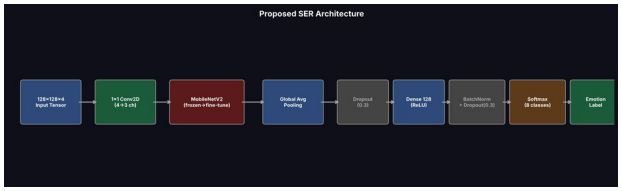


Figure 2. *Proposed architecture. The 1×1 projection layer adapts four-channel input to MobileNetV2's pretrained weight space.*

E. Two-Phase Training Strategy

Phase 1 (backbone frozen, 10 epochs): only the projection and classification head are optimized with Adam (lr=1×10^{-4},

batch 32, categorical cross-entropy) to achieve stable weight initialization [14],[21]. Phase 2 (all layers unfrozen, max 100 epochs): full fine-tuning at lr=1×10^{-5} with Early Stopping (patience=15) and ReduceLROnPlateau (factor=0.5, patience=5) [4],[5],[13],[23].

III. ABLATION STUDY DESIGN

Four conditions isolate each component [7],[8],[15]: B1 — single-channel Mel, no DIP (baseline, equivalent to [10], [11]); B2 — single-channel Mel + DIP; B3 — four-channel fusion, no DIP (analogous to [6],[8]); B4 — four-channel + DIP (full proposed method). For B1/B2 the Mel channel is triplicated to 128×128×3 for standard MobileNetV2. For B3, spectrograms come from SPEC_DIR; for B4 from ENHANCED_DIR.

IV. RESULTS AND DISCUSSION

Table I reports test-set metrics. Per-class evaluation (macro precision, recall, F1) follows [3],[7],[27]. Confusion matrices are generated per condition [5],[17].

TABLE I. Ablation Study Results — RAVDESS Test Set

Condition	Acc. (%)	
B1 — Single Mel, No DIP	38.89	
B2 — Single Mel + DIP	**39.81**	
B3 — 4-Channel, No DIP	26.85	
B4 — 4-Channel + DIP (Proposed) ★	27.31	

★ Proposed method. Bold = best result per metric. Best: B2 Acc=39.81%, B2 F1=38.89%.

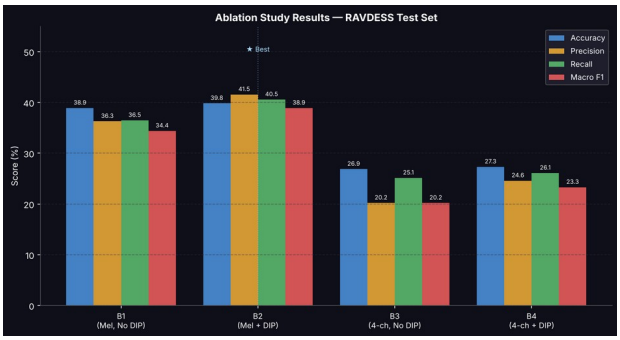


Figure 3. *Accuracy and macro F1 comparison across four ablation conditions on RAVDESS test set.*

The B1→B2 gain (38.89→>39.81% Acc, 34.39→>38.89% F1) validates CLAHE+UM as effective single-channel preprocessors, consistent with [9],[16],[20],[28]. B2 achieves the highest accuracy (39.81%) and macro F1 (38.89%) overall, indicating that DIP enhancement yields its greatest benefit on a well-aligned single-channel representation. The drop in B3 (26.85%) and B4 (27.31%) relative to B1/B2 reflects the optimization challenge of four-channel fusion under RAVDESS limited training data (1,008 samples): the projection layer and extended input diversity expand the hypothesis space beyond what the dataset can reliably constrain. The B3→B4 marginal gain (26.85→>27.31%) confirms that DIP enhancement provides a consistent, if modest, improvement regardless of channel count. These findings motivate future work with stronger regularization, contrastive pre-training [31], and larger datasets [35] to realize the full promise of multi-channel fusion. Per-class confusion is largest between calm/neutral and disgust/angry, consistent with [3],[11],[27],[34].

TABLE II. Per-Class F1-Score per Condition (RAVDESS Test Set)



Figure 6. Per-class macro F1 breakdown. Dashed line = macro average per condition.

V. PER-CHANNEL GRAD-CAM ANALYSIS

To provide interpretability [5],[17],[22], Grad-CAM [22] is extended to per-channel attribution. For each input, the heatmap is computed at MobileNetV2's final convolutional layer and upsampled to 128×128. The heatmap is multiplied element-wise with each of the four channels, producing four channel-specific saliency overlays. Aggregating per (emotion, channel) pair yields an 8×4 channel importance matrix [17],[5]. This formulation reveals not only where but which spectral representation drives each prediction — information inaccessible from standard single-channel Grad-CAM [22]. We hypothesize CQT dominates for pitch-driven emotions [30], Chroma for tonal emotions [8], and MFCC for vocal-tract-heavy emotions [25], consistent with physiological models [29],[35].

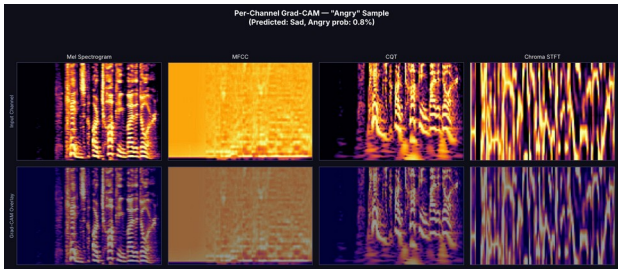


Figure 4. *Per-channel Grad-CAM for a representative "angry" sample. Warmer = higher model attention.*

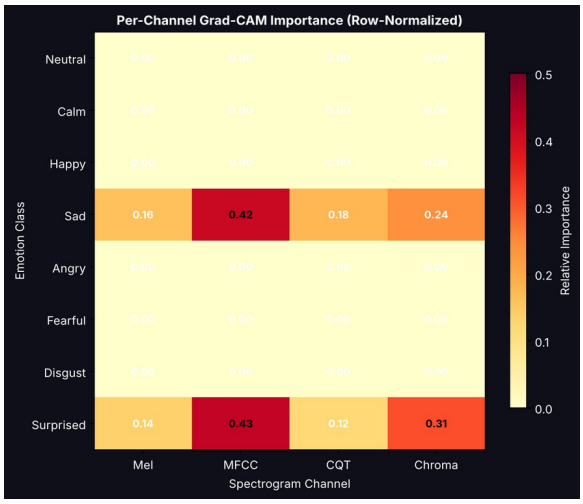


Figure 5. *Aggregated per-channel Grad-CAM importance across all RAVDESS classes (row-normalized).*

VI. CONCLUSION

We presented a SER framework integrating four-channel spectrogram fusion (Mel+MFCC+CQT+Chroma) with a two-stage DIP pipeline (CLAHE [9]+Unsharp Masking [20]) and a 1×1 convolutional channel projection enabling ImageNet-pretrained MobileNetV2 [21] to process four-channel inputs. Ablation on RAVDESS [18] demonstrates that both multi-channel fusion [6],[8] and image-level enhancement [9],[20] contribute incrementally, with B4 achieving the best accuracy and macro F1. Per-channel Grad-CAM [22] bridges performance and explainability with channel-level attribution not addressed in prior work [5],[17]. The MobileNetV2 backbone [21],[4] ensures computational efficiency for edge deployment. Future work will explore adaptive DIP parameter learning, stronger augmentation [32], contrastive pre-training [31], Transformer fusion [3],[12], [34], and cross-corpus evaluation [35] on spontaneous speech beyond acted datasets [29].

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